

HINTS AND SOLUTIONS

Q.1 C

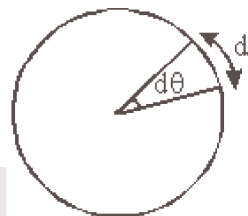
Sol. $2T \sin \frac{d\theta}{2} dm a_R$
 $= dm R \omega_0^2$... (A)

Mass per unit length = $\frac{\text{mass}}{\text{length}}$

$dm = \frac{m}{2\pi} d\theta$

By equation (A) $2T \sin \frac{d\theta}{2} = dm R \omega_0^2$

$T = \frac{m R \omega_0^2}{2\pi} \left\{ \sin \frac{d\theta}{2} = \frac{d\theta}{2} \right\}$



Q.2 D

Q.3 B

Sol. The total K.E. at B = P.E. at A; only the translational K.E. is responsible for the sphere gaining height during its motion along BC.

Q.4 A

Sol. Linear and angular momentum of the system are conserved.

Q.5 B

Sol. Solve the problem using μ as the coefficient of kinetic friction.

Q.6 B

Sol. Net force = mass $\times a_{cm}$,
 the center of mass has both tangential and radial accelerations

Q.7 D

Sol. $W = \frac{1}{2} I \omega^2$

$I = m_1 x_1^2 + m_2 (\ell - x)^2$

I should be minimum.

Q.8 A

Sol. $L_{A/B} = mv_0 \ell - m a \ell$

Q.9 C

Sol. $a_{cm} = \frac{F}{M}$ and $\alpha = \frac{T}{I} = \frac{FR}{MR^2} = \frac{F}{MR}$

$a_{cm} = \alpha R$

So, the body will roll without slipping due to applied force F hence friction will be absent.

Q.10 B

Sol. (b) The centre of gravity of the pipe lies at a distance $\left(\frac{2r}{\pi}\right)$

Q.11 C

Sol. $a_1 = R \alpha - a_2$
 $2 = 2\alpha - 4$
 $\alpha = 3 \text{ rad/s}^2$.

Q.12 B

Sol. $v_2 = \omega \ell \sin \theta$

$$\omega = \frac{v_2}{\ell \sin \theta}$$

Q.13 A

Sol. The angular momentum is $I\omega$,

$$\text{where } I = \frac{1}{3} mL^2 \sin^2 \theta$$

$$\text{and } \omega = \frac{v}{L \sin \theta}$$

Q.14 B

Sol. For equilibrium c.m. of the ring must lie on vertical line.

Q.15 B

Sol. $F + f = 2ma$ (Newton second law) ... (i)

Torque about (0, 0)

$$\tau_0 = FR - fR = \frac{4}{3} mR^2 \frac{a}{R} \quad \dots (ii)$$

Solving equation (i) and (ii)

$$a = \frac{3F}{5m}$$

Q.16 C

Sol. Torque about point A

$$F \cdot \frac{a\sqrt{3}}{2} = mg \cdot \frac{a}{2}$$

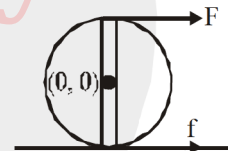
$$F = \frac{mg}{\sqrt{3}} = 0.571 mg$$

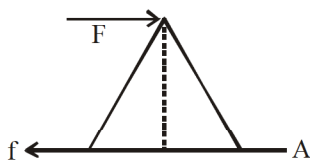
for equilibrium of block

$$F - f = 0$$

$$F = f$$

μ should be greater than 0.577.





- Q.17 A
Q.18 D
Q.19 C
Q.20 AC

Sol. (A) $E = \frac{1}{2} \times m \left(\frac{l}{2} \right)^2 \omega^2 \times 2$

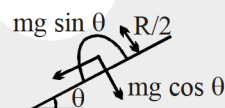
(B) $T_1 \sin 60^\circ = mg$
 $T_1 \cos 60^\circ + T_2 = m\omega^2 l/2$
 $\frac{mg}{\sqrt{3}} + T_2 = m\omega^2 \frac{l}{2}$

$$T_2 = m\omega^2 \frac{l}{2} - \frac{mg}{\sqrt{3}}$$

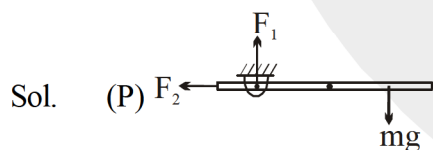
(C) $T_1 = \frac{2mg}{\sqrt{3}}$

- Q.21 BD
Q.22 ABC
Q.23 ABC

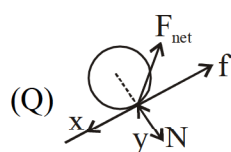
Sol. $mg \sin \theta \frac{R}{2} \geq mg \cos \theta R$
 $\tan \theta \geq 2$
 $mg \sin \theta \leq \mu mg \cos \theta$
 $\mu \geq \tan \theta$



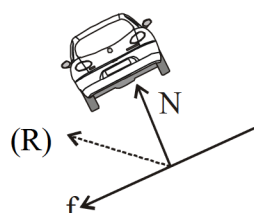
- Q.24 [Ans. (A) R (B) P (C) QT (D) ST]



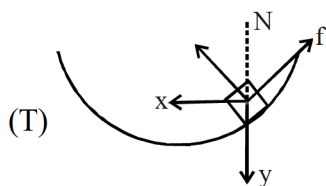
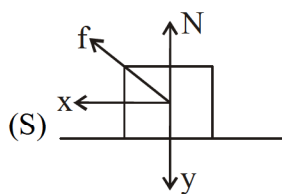
f_2 provides centripetal acceleration.



f provides the torque to roll



f & N provides centripetal force



Q.25 [Ans. 6 m/s]

Sol. By conservation of angular momentum.

$$mv_0 R = mvR + \frac{2}{3}mvR$$

$$\frac{5}{3}mvR$$

$$v = \frac{3v_0}{5} = 6 \text{ m/s}$$

Rankers Academy JEE